

Lecture 1

EE-215 Electronic Devices & Circuits

Asst Prof Muhammad Anis Ch

EE-215 Electronic Devices & Circuits

- Credit Hours: 3-1
- Course Book:
 - Adel S. Sedra and Kenneth C. Smith, "Microelectronic Circuits", Sixth Edition
- Reference Book:
 - B. Razavi, "Fundamentals of Microelectronics", Second Edition
 - B.G. Streetman and S. Banerjee, "Solid State Electronic Devices", Sixth Edition
- Software Tools
 - LTspice (visit www.linear.com for LTspice "setup" and "Getting Started Guide")
 - Matlab
- Our emphasis in this course is on semiconductor devices based circuits.

Semiconductor Devices:

- Diodes
- Bipolar Junction Transistors (BJT)
- Field Effect Transistors (FET)
 - the semiconductor electronic devices can be used singly along with resistors, capacitors and inductors
 - for the design of discrete circuits.
 - Or as components of an integrated circuit (IC) chip
- Integrated Circuit(IC)
 - An integrated circuit (IC) has multiple electronic devices (diodes or transistors) in a small piece of silicon (or any other semiconductor).
 - e.g. A microprocessor has hundreds of millions of components in a small piece of silicon whose area is on the order of 100mm^2

Diodes

- A Diode
 - is a two terminal device
 - is the most fundamental non-linear circuit element

The Ideal Diode

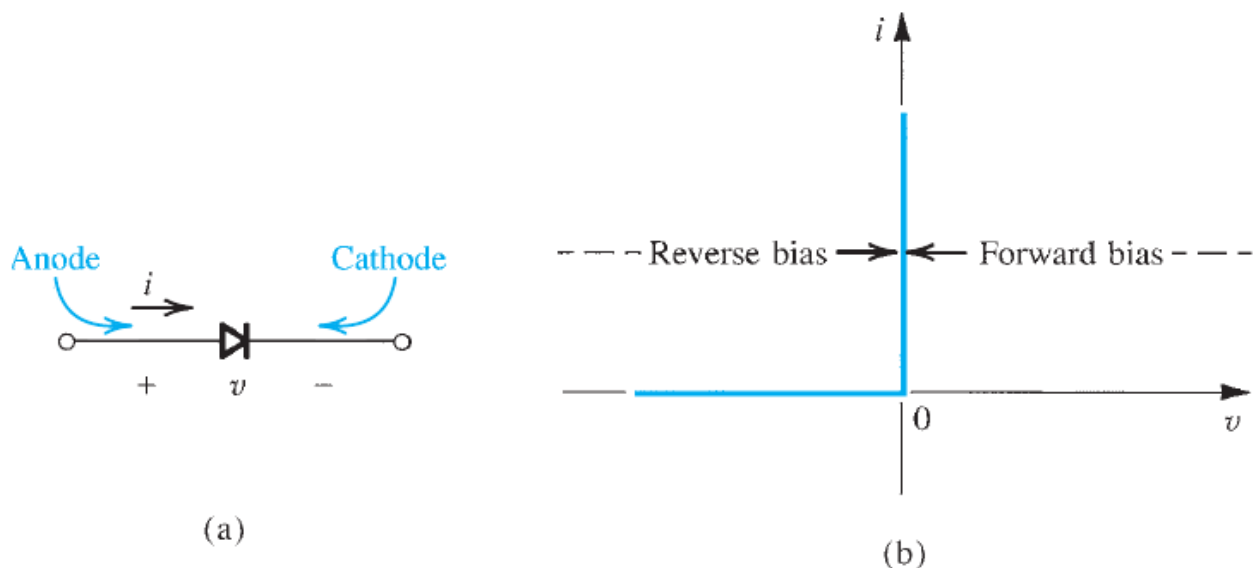


Figure 4.1 The ideal diode: (a) diode circuit symbol; (b) $i-v$ characteristic;

- from the $i-v$ characteristics of the ideal diode,
 - one can see that the diode can operate in two distinct modes
 - either as a Reverse Biased Diode
 - or as a Forward Biased Diode

Reverse Biased Diode

- If a negative voltage ' v ' is applied to the diode,
 - no current flows through the diode
 - the diode behaves as an open circuit
 - and the diode is said to be cutoff

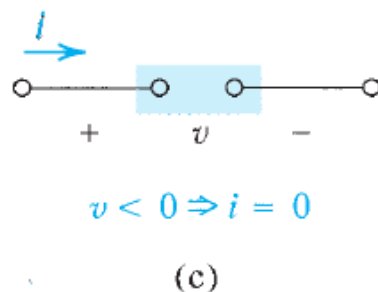
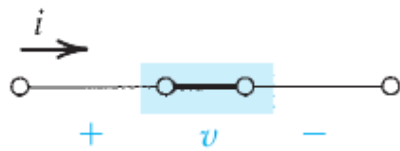


Figure 4.1 The ideal diode: (c) equivalent circuit in the reverse direction;

Forward Biased Diode

- If a positive current i is applied to the ideal diode,
 - zero voltage drop appears across the diode.
 - i.e. the diode acts as a short circuit in the forward direction
 - the diode will pass any current with zero voltage drop.



$$i > 0 \Rightarrow v = 0$$

(d)

Figure 4.1 The ideal diode:
(d) equivalent circuit in the forward direction.

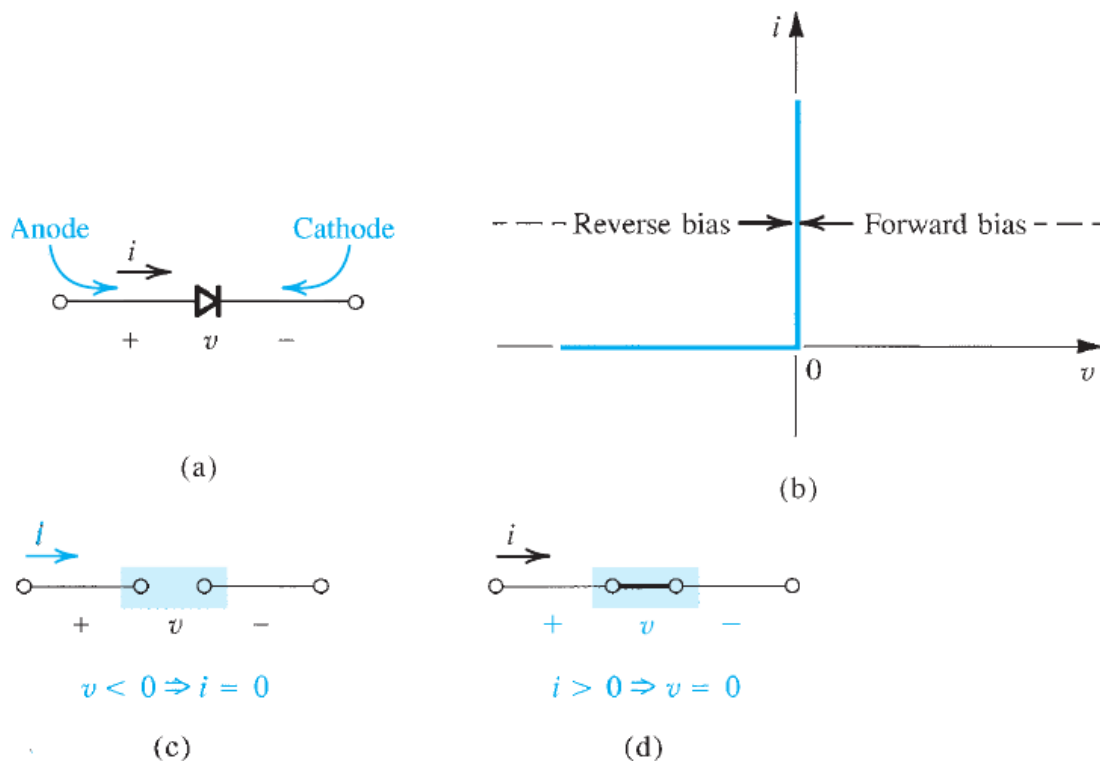


Figure 4.1 The ideal diode: (a) diode circuit symbol; (b) i - v characteristic; (c) equivalent circuit in the reverse direction; (d) equivalent circuit in the forward direction.

- The external circuit must be designed to limit
 - the current through the conducting (forward biased) diode
 - the reverse voltage across a cut off (reverse biased) diode

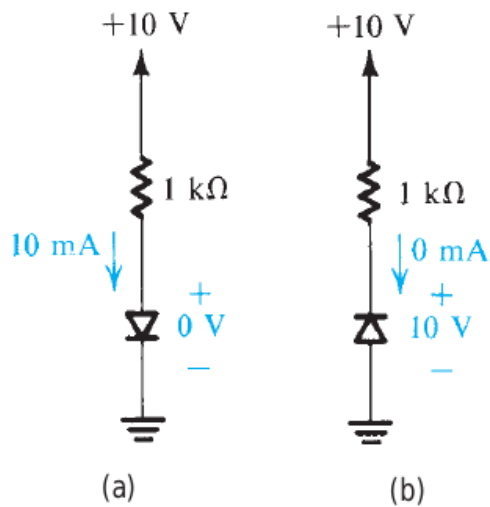


Figure 4.2 The two modes of operation of ideal diodes and the use of an external circuit to limit **(a)** the forward current and **(b)** the reverse voltage.

- The external circuit must be designed to limit
 - the current through the conducting (forward biased) diode (10mA in fig-a)
 - the reverse voltage across a cut off (reverse biased) diode (10V in fig-b)

A Simple Application: The Rectifier

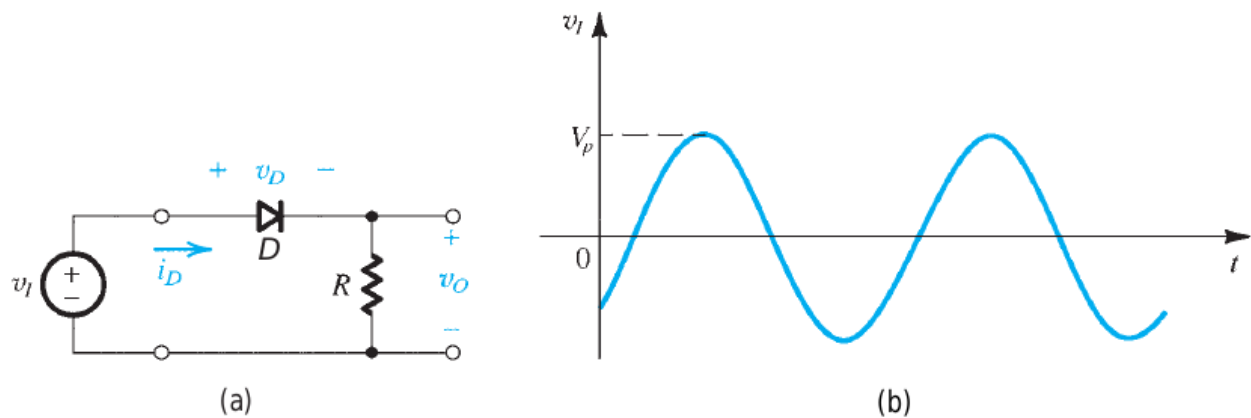
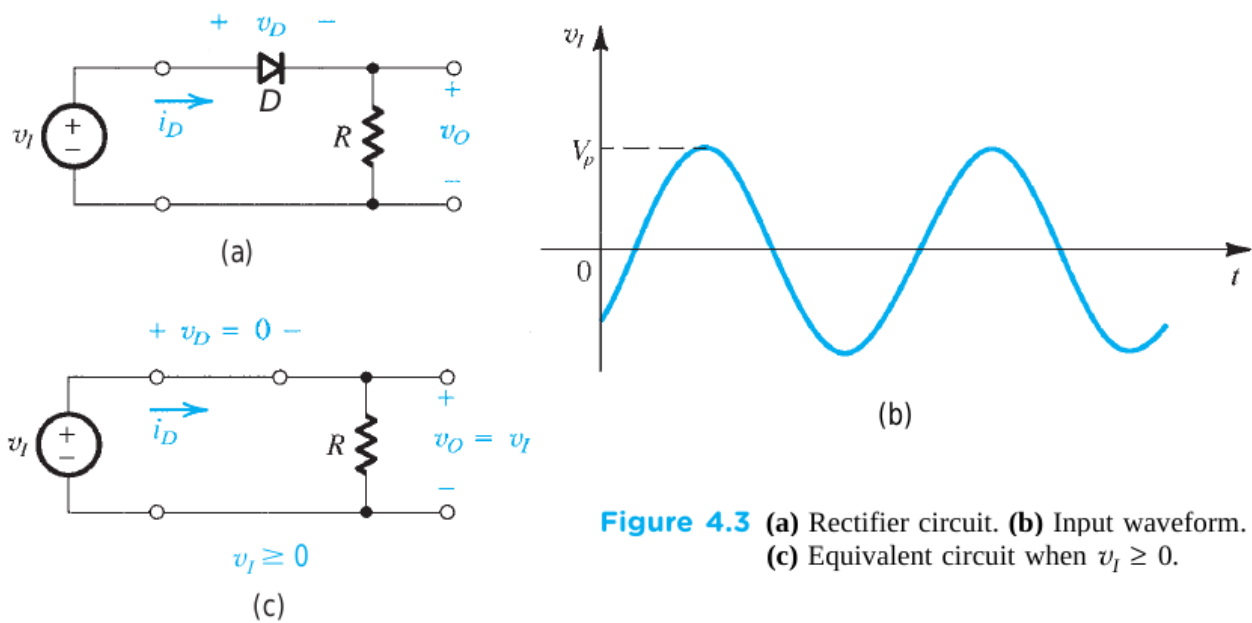


Figure 4.3 (a) Rectifier circuit. (b) Input waveform.

- A diode can be used as a rectifier
 - i.e. to convert AC to DC
 - A simple circuit consisting of series connection of a diode and a resistor, can act as a rectifier.
 - the diode is assumed to be ideal and the input signal is a sinusoid.



- Now during the positive half-cycle of the input sinusoid,
 - +ve v_I will cause current i_D to flow through the diode
 - i.e. the diode is forward biased
 - $\Rightarrow v_D$, the voltage across the diode will be zero
- Now during the negative half-cycle of the input sinusoid,
 - a -ve voltage v_D will appear across the diode
 - hence the diode is reverse biased
 - and will act as an open circuit
- As a consequence, the output voltage waveform will be of the form

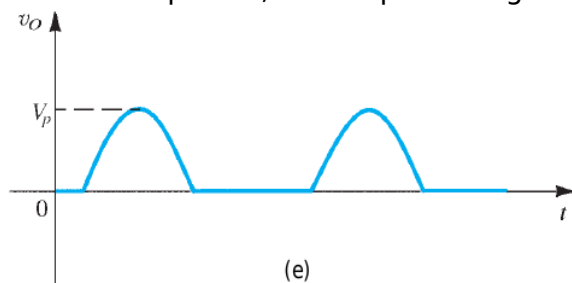


Figure 4.3 (e) Output waveform.

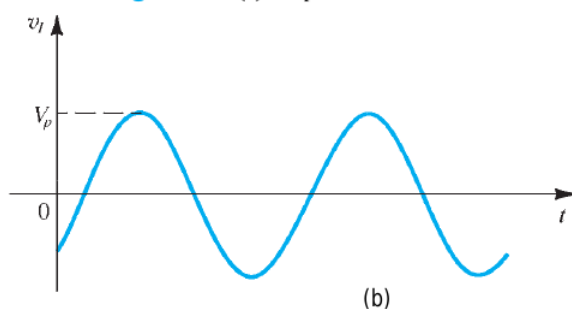


Figure 4.3 (b) Input waveform.

- Note that
- v_I alternated in polarity and has a zero average value.
- while v_O is unidirectional and has a finite average value (or a DC component)

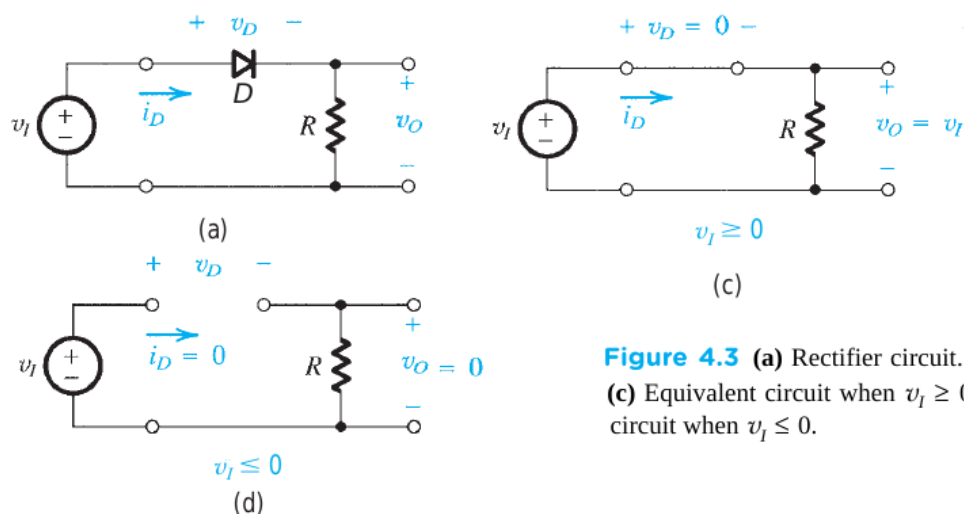


Figure 4.3 (a) Rectifier circuit. (c) Equivalent circuit when $v_I \geq 0$. (d) Equivalent circuit when $v_I \leq 0$.

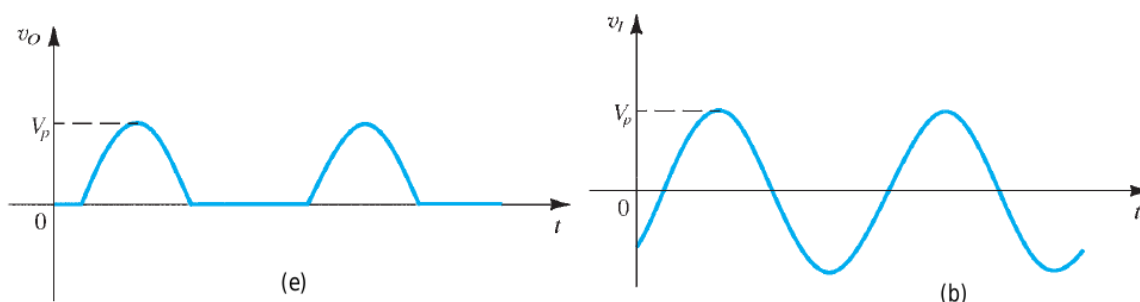


Figure 4.3 (e) Output waveform.

Figure 4.3 (b) Input waveform.

Exercise 4.2

- For the circuit in Fig. 4.3(a), sketch the waveform of v_D

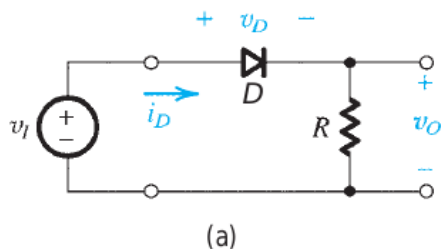


Figure 4.3 (a) Rectifier circuit.

Solution

- from the circuit, v_D can be given as $v_D = v_I - v_O$

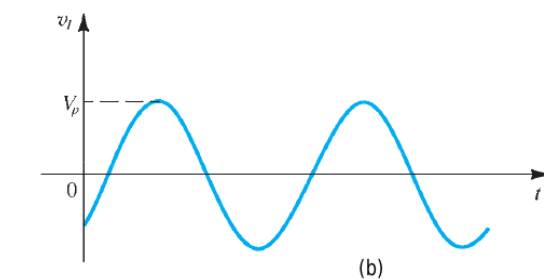


Figure 4.3 (b) Input waveform.

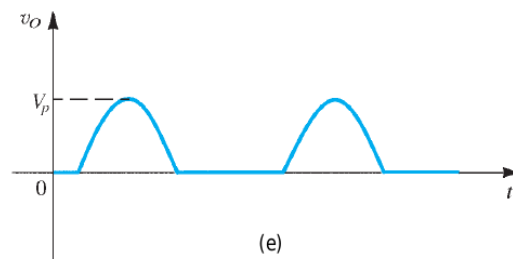


Figure 4.3 (e) Output waveform.

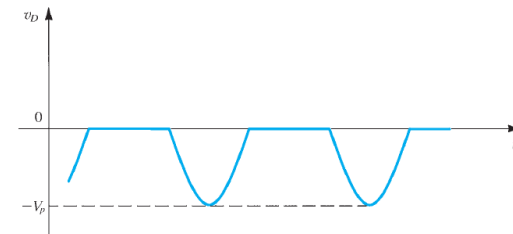


Figure E4.2

Exercise 4.3

- In the circuit of Fig. 4.3(a), let v_I have a peak value of 10V and $R=1k\Omega$. Find the peak value of i_D and the dc component of v_O .

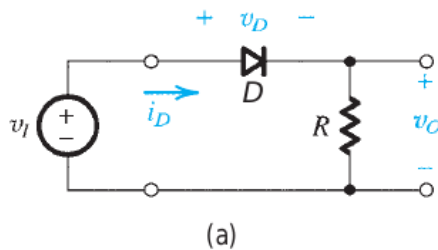


Figure 4.3 (a) Rectifier circuit.

Solution

- peak value of $v_I = V_P = 10V$
 - peak value of i_D will occur when $v_I = V_P$ and diode is conducting
 - peak value of $i_D = I_P = \frac{V_P}{R} = \frac{10}{1k} = 10mA$

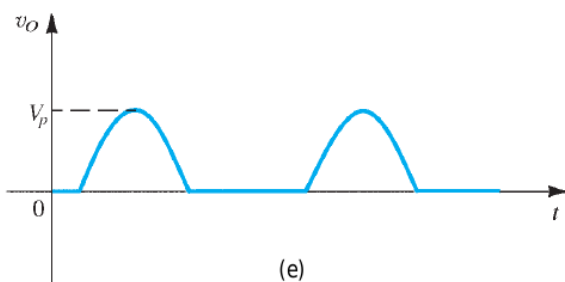


Figure 4.3 (e) Output waveform.

- the average value (or the dc component) of v_O can be given as
 - $v_{O,Average} = \frac{1}{T} \int_0^T v_O(t) dt$
 - from the figure, $v_O(t) = \begin{cases} V_P \sin \omega t & \text{for } 0 \leq t \leq \frac{T}{2} \\ 0 & \text{for } \frac{T}{2} \leq t \leq T \end{cases}$.

$$\circ \Rightarrow v_{O,Average} = \frac{1}{T} \left[\int_0^{T/2} v_O(t) dt + \int_{T/2}^T v_O(t) dt \right] = \frac{1}{T} \left[\int_0^{T/2} V_P \sin \omega t dt \right]$$

$$\bullet v_{O,Average} = \frac{1}{T} \left[\int_0^{T/2} V_P \sin \omega t dt \right]$$

$$\circ v_{O,Average} = \frac{V_P}{T} \left[\int_0^{T/2} \sin \omega t dt \right] = \frac{V_P}{T} \left[\frac{-\cos \omega t}{\omega} \right]_0^{T/2} = \frac{V_P}{T\omega} [-\cos \omega t]_0^{T/2}$$

$$\circ \text{ As } \omega = \frac{2\pi}{T} \text{ and } \omega T = 2\pi$$

$$\circ v_{O,Average} = \frac{V_P}{2\pi} [-\cos \frac{2\pi}{T} t]_0^{T/2} = \frac{V_P}{2\pi} \left[\left(-\cos \frac{2\pi}{T} \frac{T}{2} \right) - \left(-\cos \frac{2\pi}{T} 0 \right) \right]$$

$$\circ v_{O,Average} = \frac{V_P}{2\pi} [(-\cos \pi) + \cos 0] = \frac{V_P}{2\pi} [1 + 1] = \frac{V_P}{\pi}$$

$$\bullet \text{ thus for } V_P = 10V$$

$$\circ \Rightarrow v_{O,Average} = \frac{V_P}{\pi} = \frac{10}{\pi} = 3.1831V$$

Another Application: Diode Logic Gates

- Diodes along with resistors can be used to
 - implement digital logic gates.
 - Truth tables for two input AND gate and a two input OR gate can be given as

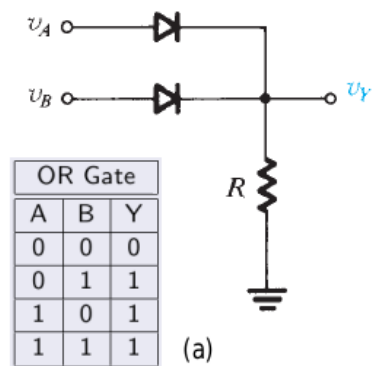
OR Gate			AND Gate		
A	B	Y	A	B	Y
0	0	0	0	0	0
0	1	1	0	1	0
1	0	1	1	0	0
1	1	1	1	1	1

- where

- logic 0 corresponds to a voltage value near 0V
- and logic 1 corresponds to a voltage value near +5V

OR Gate

- when $v_A = v_B = +5V = \text{logic 1}$
 - Both diodes are forward biased (conducting)
 - $\Rightarrow v_Y = +5V$
- when $v_A = 0V$, $v_B = +5V$,
 - Diode A is cut off, Diode B is conducting
 - $\Rightarrow v_Y = +5V$
- when $v_A = +5V$, $v_B = 0V$,
 - Diode A is conducting, Diode B is off
 - $\Rightarrow v_Y = +5V$

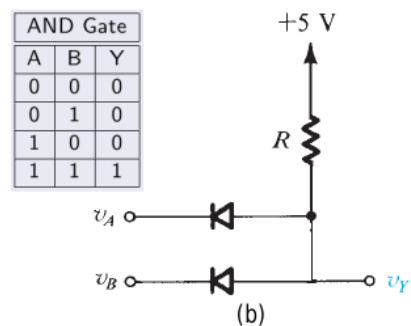


■ Diode logic gates: (a) OR gate;

- when $v_A = v_B = 0V = \text{logic}0$, Both diodes are reverse biased, $v_Y = 0V$

AND Gate

- when $v_A = v_B = 0V = \text{logic}0$
 - Both diodes are forward biased (conducting)
 - $\Rightarrow v_Y = 0V$
 - when $v_A = 0V, v_B = +5V$,
 - Diode A is conducting, Diode B is cut off
 - $\Rightarrow v_Y = 0V$
 - when $v_A = +5V, v_B = 0V$,
 - Diode A is off, Diode B is conducting
 - $\Rightarrow v_Y = 0V$



- (b) AND gate (in a positive-logic system).
- when $v_A = v_B = 0V = \text{logic}0$, Both diodes are conducting, $v_Y = 0V$
- when $v_A = v_B = +5V = \text{logic}1$, Both diodes are reverse biased (cutoff), $v_Y = +5V$

- Similarly 3 input AND, OR functions can be implemented using 3 diodes.

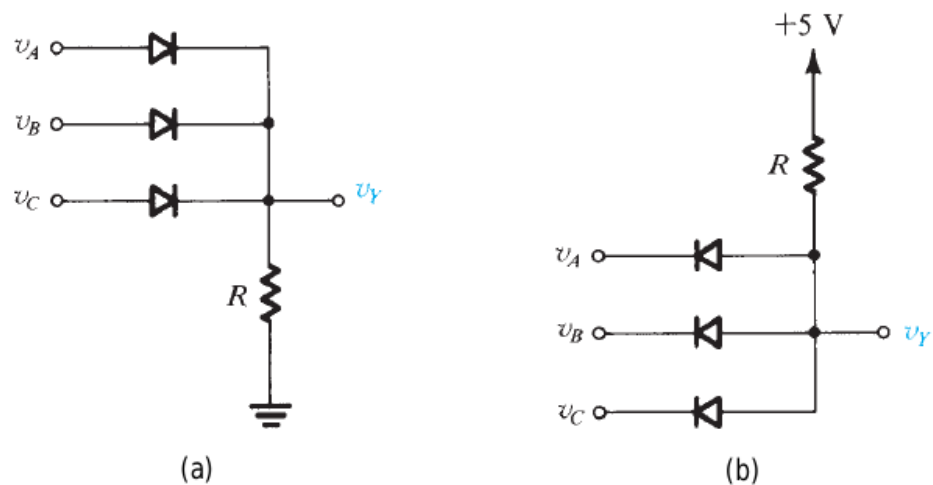


Figure 4.5 Diode logic gates: (a) OR gate; (b) AND gate (in a positive-logic system).

Example 4.2

- Assuming the diodes to be ideal, find the values of I and V , in the circuits of Fig. 4.6.

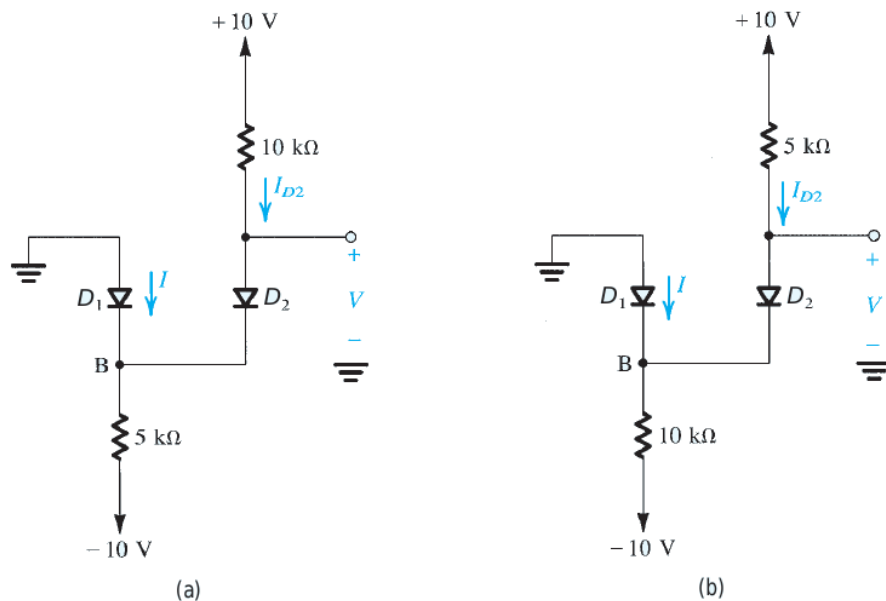


Figure 4.6 Circuits for Example 4.2.

- Note:
 - It is not always clear in a circuit,
 - whether the diode is conducting or not
- In such a case,
 - make an assumption
 - proceed with the analysis
 - see if the solution is reasonable
- e.g. if assumption is that the diode is conducting,
 - its current i_D must be positive.
- for circuit (a), lets assume both diodes are conducting
 - $\Rightarrow$ we can replace both diodes with short circuits
 - $\Rightarrow V_B = 0V$ and $V = 0V$

- Applying ohms' law across 10k resistor

$$\circ I_{D2} = \frac{10-0}{10k} = 1mA$$

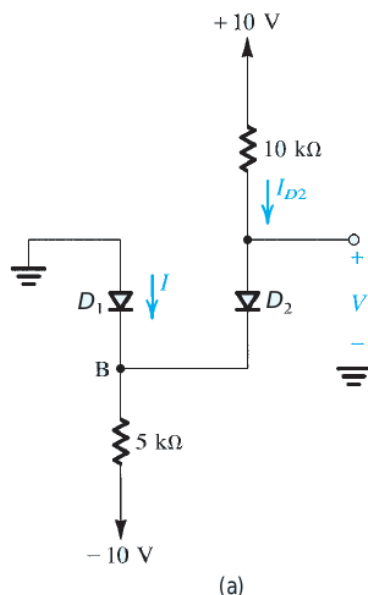
- KCL at node formed by V and V_B (0V)

$$\circ I_{D2} + I = I_{5k(Resistor)} = \frac{V_B - (-10)}{5k} = \frac{0 - (-10)}{5k} = \frac{10}{5k} = 2mA$$

$$\circ \Rightarrow I = 2mA - I_{D2} = 1mA$$

- As I and I_{D2} both are positive currents

- our assumption was correct i.e. both diodes are conducting.



■ **Figure 4.6** Circuits for Example 4.2.

for circuit (b),

- lets assume both diodes are conducting

- $\Rightarrow$ we can replace both diodes with short circuits

- $\Rightarrow V_B = 0V$ and $V = 0V$

- Applying ohms' law across 5k resistor

$$\circ I_{D2} = \frac{10-0}{5k} = 2mA$$

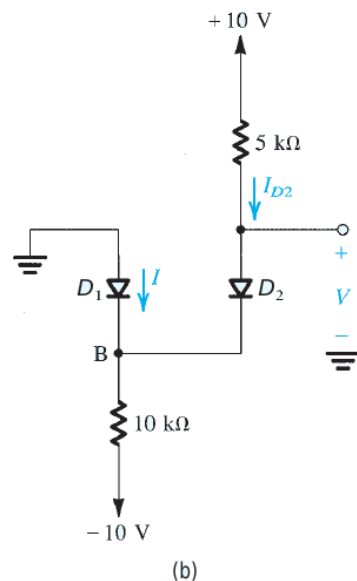
- KCL at node formed by V and V_B (0V)

$$\circ I_{D2} + I = I_{10k(Resistor)} = \frac{V_B - (-10)}{10k} = \frac{0+10}{10k} = \frac{10}{10k} = 1mA$$

$$\circ \Rightarrow I = 1mA - I_{D2} = -1mA$$

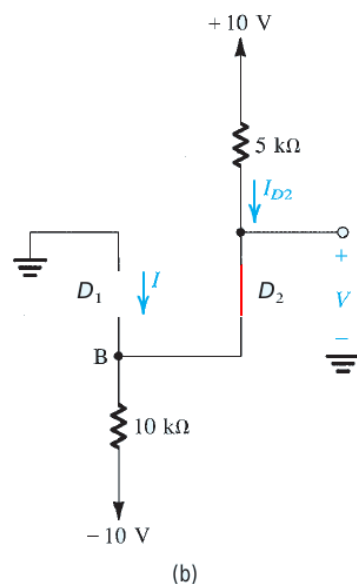
- As I_{D2} is positive but I is negative

- $\Rightarrow D_2$ is conducting but D_1 is cut off



■ **Figure 4.6** Circuits for Example 4.2.

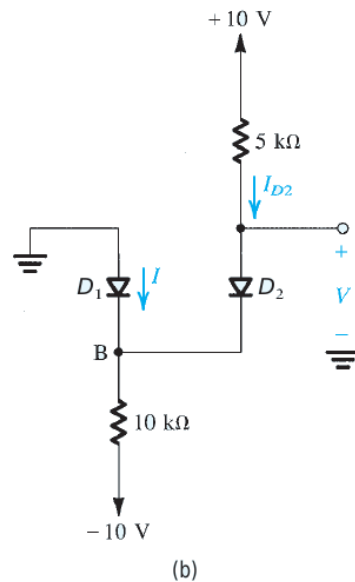
- Now, D_2 is conducting but D_1 is cut off
 - $\Rightarrow$ we can replace D_2 with short circuit, D_1 with open circuit
- As D_1 is cut off $\Rightarrow I = 0$
 - I_{D2} is the current flowing through series combination of 5k and 10k resistors
- Applying ohms' law across (5k+10k) resistor
 - $I_{D2} = \frac{10 - (-10)}{5k + 10k} = \frac{20}{15k} = 1.33mA$
- Ohm's law across 5k resistor
 - $\frac{10 - V}{5k} = I_{D2} = 1.33mA$
 - $\Rightarrow 10 - V = 6.65V$
 - or $V = 3.35V$



■ **Figure 4.6** Circuits for Example 4.2.

- $I_{D2} = 1.33mA$ and $V = 3.35V$
 - $\Rightarrow I_{D2}$ is +ve $\Rightarrow D2$ is conducting
- Note that $V_B = V = 3.35V$

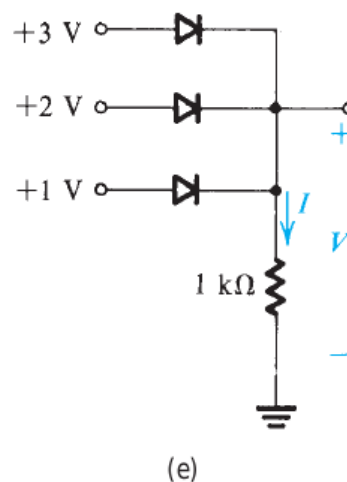
- $\Rightarrow V_{D1} = -3.35V$
- thus D_1 is reverse biased



■ **Figure 4.6** Circuits for Example 4.2.

Exercise 4.4e

- Find the values of I and V in the circuits shown in Fig. E4.4e.
 - only +3V diode conducting
 - other two diodes off
 - $V = +3V$
 - $I = \frac{3}{1k} = 3mA$
 - this circuit can be used to select one of the 3 DC sources

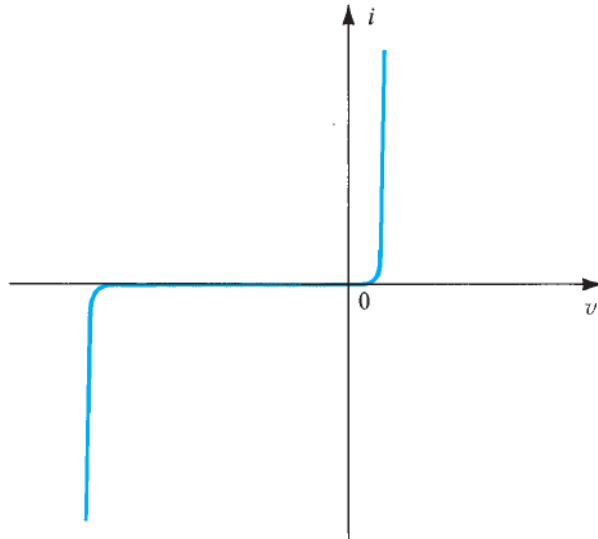


■ **Figure E4.4** (Continued)

Terminal Characteristics of Junction Diodes

- Uptill now we have considered
 - the behaviour of ideal diode
 - lets take a look at the characteristics of real diodes
 - specifically, semiconductor junction diodes made of Silicon

- o i-v characteristics of a silicon PN junction diode is shown



- o **Figure 4.7** The $i-v$ characteristic of a silicon junction diode.

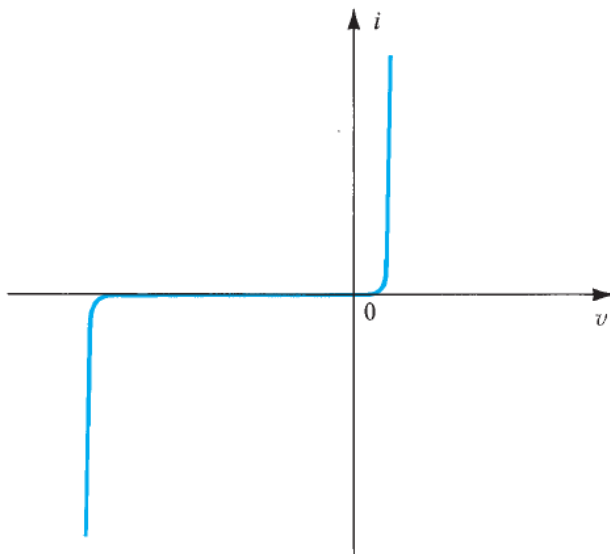


Figure 4.7 The $i-v$ characteristic of a silicon junction diode.

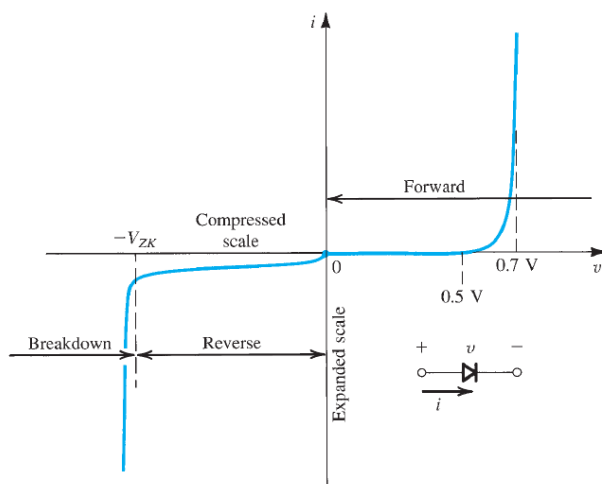


Figure 4.8 The diode $i-v$ relationship with some scales expanded and others compressed in order to reveal details.

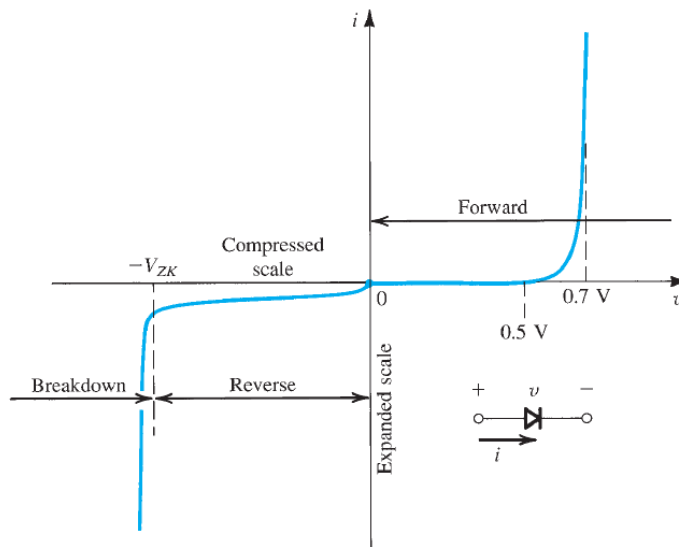


Figure 4.8 The diode i - v relationship with some scales expanded and others compressed in order to reveal details.

- the characteristic curve consists of three distinct regions
 - The forward bias region, determined by $v > 0$
 - The reverse bias region, determined by $v < 0$
 - The breakdown region, determined by $v < -V_{zk}$
- A diode is in forward-bias region of operation
 - when the terminal voltage v is positive
 - In the forward bias region, i - v relationship can be given as
 - $i = I_S \left(e^{\frac{v}{nV_T}} - 1 \right)$
 - where $I_S = \text{Saturation Current}$
 - I_S is also called scale current.
 - As I_S is directly proportional to the cross-sectional area of the diode.
 - Thus doubling of the junction area $\Rightarrow$ doubling of I_S
 - for small-signal diodes, I_S is on the order of 10^{-15} A
 - value of I_S is a very strong function of temperature
 - As a rule of thumb, I_S doubles for every 5°C rise in temperature
 - n is called ideality factor. $1 \leq n \leq 2$. Assume $n = 1$

The Forward-Bias Region

- $i = I_S \left(e^{\frac{v}{V_T}} - 1 \right)$
 - where V_T is a constant called thermal voltage
 - and is $V_T = \frac{kT}{q} = 0.0862T \text{ mV}$
 - $T = \text{absolute temperature in kelvins} = 273 + \text{temperature in } ^\circ \text{C}$
 - At room temperature (20°C),
 - $V_T = 0.0862T \text{ mV} = 0.0862(293) \text{ mV} = 25.26 \text{ mV}$
 - for ease of calculations,

$$\blacksquare V_T \approx 25 \text{ mV at room temperature}$$

$$\bullet i = I_S \left(e^{\frac{v}{V_T}} - 1 \right)$$

$$\circ \Rightarrow i = I_S e^{\frac{v}{V_T}} - I_S$$

$$\blacksquare \Rightarrow i + I_S = I_S e^{\frac{v}{V_T}}$$

◦ for $i > I_S$, which is normally the case for a forward-bias diode

$$\blacksquare i + I_S = I_S e^{\frac{v}{V_T}} \Rightarrow i \approx I_S e^{\frac{v}{V_T}}$$

$$\circ i \approx I_S e^{\frac{v}{V_T}}$$

$$\circ \text{ or } \frac{i}{I_S} \approx e^{\frac{v}{V_T}}$$

◦ or in logarithmic form

▪ taking natural log (base e) on both sides

$$\blacksquare \ln\left(\frac{i}{I_S}\right) \approx \ln\left(e^{\frac{v}{V_T}}\right)$$

$$\blacksquare \ln\left(\frac{i}{I_S}\right) \approx \frac{v}{V_T} \ln(e) = \frac{v}{V_T} \Rightarrow v \approx V_T \ln\left(\frac{i}{I_S}\right)$$

• two forms of i-v relationship for the forward-bias region

$$\circ i \approx I_S e^{\frac{v}{V_T}} \text{ and } v \approx V_T \ln\left(\frac{i}{I_S}\right)$$

• if now diode voltage is V_1 ,

$$\circ \text{ its current } I_1 \text{ can be given as } I_1 = I_S e^{\frac{V_1}{V_T}} \dots\dots\dots(\text{A})$$

◦ if diode voltage is changed to V_2 ,

$$\blacksquare \text{ then its current } I_2 \text{ will be } I_2 = I_S e^{\frac{V_2}{V_T}} \dots\dots\dots(\text{B})$$

$$\circ \text{ divide (B) by (A)} \Rightarrow \frac{I_2}{I_1} = \frac{I_S e^{\frac{V_2}{V_T}}}{I_S e^{\frac{V_1}{V_T}}} = \frac{e^{\frac{V_2}{V_T}}}{e^{\frac{V_1}{V_T}}} = e^{\frac{V_2}{V_T} - \frac{V_1}{V_T}} = e^{\frac{V_2 - V_1}{V_T}}$$

$$\circ \text{ taking natural log on both sides} \Rightarrow \ln\left(\frac{I_2}{I_1}\right) = \frac{V_2 - V_1}{V_T}$$

$$\circ \text{ or } V_2 - V_1 = V_T \ln\left(\frac{I_2}{I_1}\right)$$

$$\bullet V_2 - V_1 = V_T \ln\left(\frac{I_2}{I_1}\right)$$

◦ or in terms of base-10 logarithms

▪ using the identity

$$\blacksquare \ln x = \log_e x = \frac{\log_{10} x}{\log_{10} e} = \frac{\log x}{\log e} = 2.3 \log x$$

$$\circ \Rightarrow V_2 - V_1 = 2.3 V_T \log\left(\frac{I_2}{I_1}\right)$$

▪ $\Rightarrow$ for a decade (factor of 10) change in current,

▪ the diode voltage drop changes by only $2.3 V_T = 57.5 \text{ mV}$

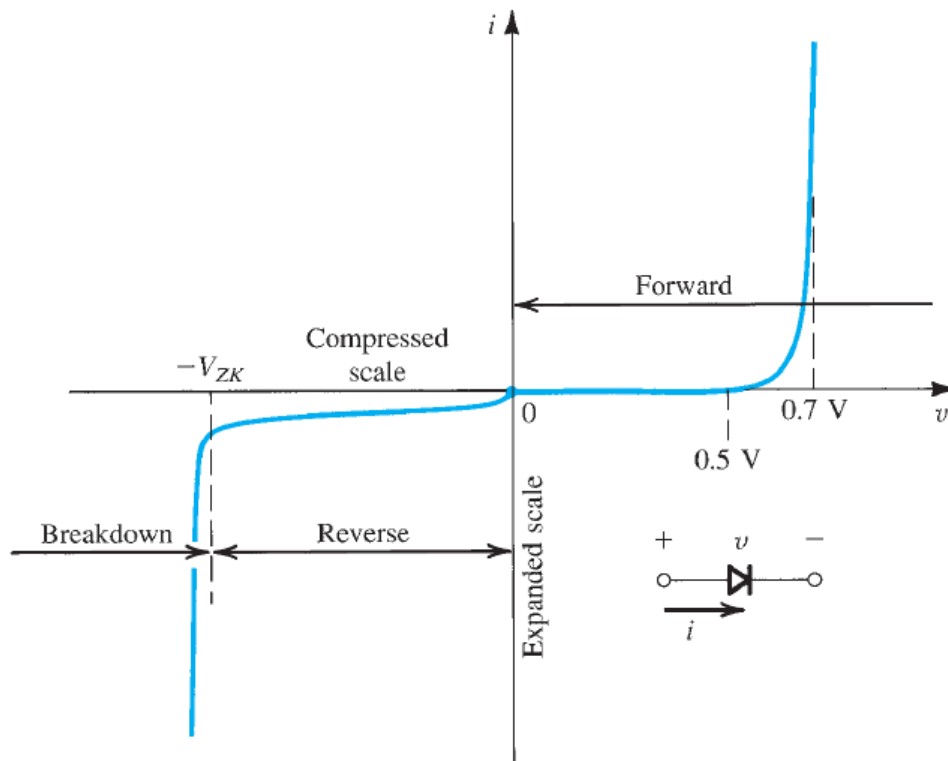


Figure 4.8 The diode i - v relationship with some scales expanded and others compressed in order to reveal details.

- from the i - v characteristics in the forward-bias region
 - The current is negligibly small for
 - v smaller than about 0.5V
 - This value is called cut-in voltage.
 - Also the current rapidly increases afterwards
 - for a fully conducting diode,
 - the voltage drop lies in a narrow range,
 - approximately 0.6V to 0.8V
 - This observation leads to a simple model for the diode
 - where it can be assumed that a conducting diode
 - has a 0.7V voltage drop across it.

Exercise 4.6

- Find the change in diode voltage if the current changes from 0.1mA to 10mA

Solution

- $I_1 = 0.1\text{mA}$
 - $I_2 = 10\text{mA}$
 - $V_2 - V_1 = V_T \ln \frac{I_2}{I_1} = 25\text{mV} \ln \frac{10}{0.1} = 0.115\text{ V}$
 - or $V_2 - V_1 = 115\text{ mV}$

Exercise 4.7

- A silicon junction diode has $v=0.7\text{ V}$ at $i=1\text{ mA}$. Find the voltage drop at $i=0.1\text{ mA}$ and $i=10\text{ mA}$.

Solution

- $V_1 = 0.7 \text{ V}$ and $I_1 = 1 \text{ mA}$
 - $V_2 - V_1 = ?$, $I_2 = 0.1 \text{ mA}$
 - $V_3 - V_1 = ?$, $I_3 = 10 \text{ mA}$
 -
 - $V_2 - V_1 = V_T \ln \frac{I_2}{I_1} = 25 \text{ mV} \ln \frac{0.1}{1} = -0.058 \text{ V}$
 - $V_2 - 0.7 = -0.058 \text{ V}$
 - $\Rightarrow V_2 = 0.7 - 0.058 = 0.642 \text{ V}$
 - $V_3 - V_1 = V_T \ln \frac{I_3}{I_1} = 25 \text{ mV} \ln \frac{10}{1} = 0.058 \text{ V}$
 - $V_3 - 0.7 = 0.058 \text{ V}$
 - $\Rightarrow V_3 = 0.7 + 0.058 = 0.758 \text{ V}$

Exercise 4.8

- Using the fact that a silicon diode has $I_S = 10^{-14} \text{ A}$ at 25°C and that I_S increases by 15% per $^\circ \text{C}$ rise in temperature, find the value of I_S at 125°C .

Solution

- $I_S = 10^{-14} \text{ A}$ at 25°C
 - I_S increases by 15% per $^\circ \text{C}$ rise in temperature
 - I_S at $125^\circ \text{C} = ?$
 - $I_{S,25^\circ \text{C}} = 10^{-14} \text{ A}$
 - $I_{S,26^\circ \text{C}} = I_{S,25^\circ \text{C}} + 0.15 I_{S,25^\circ \text{C}} = 1.15 I_{S,25^\circ \text{C}}$
 - $I_{S,27^\circ \text{C}} = I_{S,26^\circ \text{C}} + 0.15 I_{S,26^\circ \text{C}} = 1.15 I_{S,26^\circ \text{C}} = (1.15)^2 I_{S,25^\circ \text{C}}$
 - $\Rightarrow I_{S,125^\circ \text{C}} = (1.15)^{(125-25)} I_{S,25^\circ \text{C}} = (1.15)^{100} I_{S,25^\circ \text{C}}$
 - thus I_S at $125^\circ \text{C} = (1.15)^{100} \times 10^{-14} \text{ A} = 1.1743 \times 10^{-8} \text{ A}$

The Reverse-Bias Region

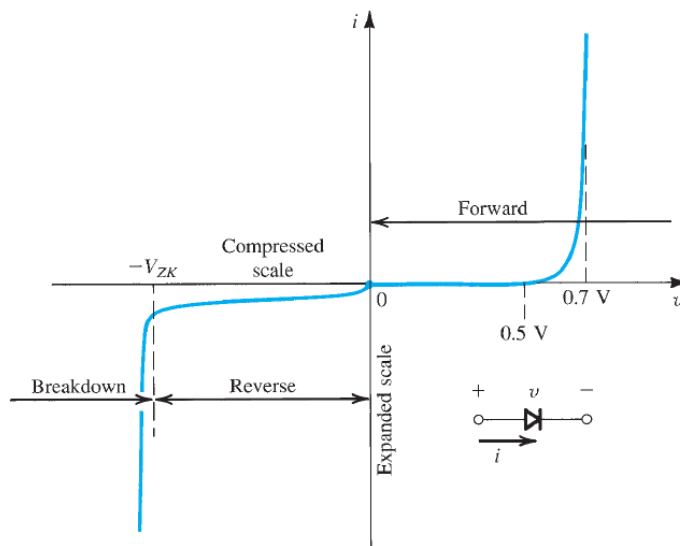
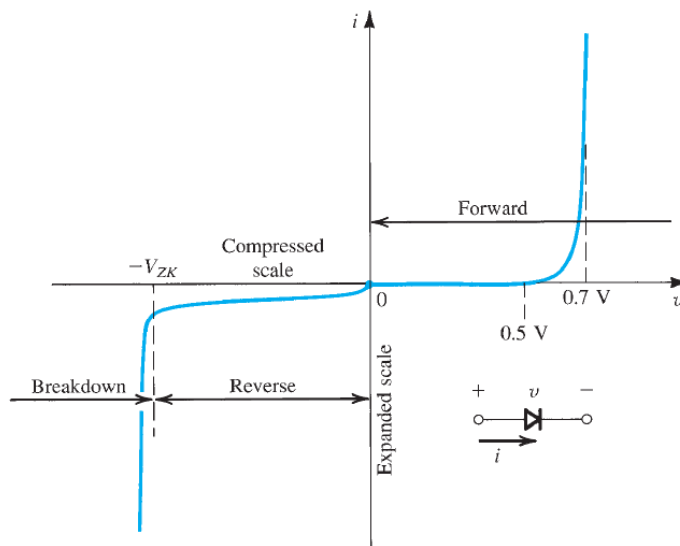


Figure 4.8 The diode i - v relationship with some scales expanded and others compressed in order to reveal details.

- The reverse-bias region of operation is entered
 - when the diode voltage v is made negative.
- As $i = I_S \left(e^{\frac{v}{V_T}} - 1 \right)$

- if v is -ve and a few times larger than $V_T = 25mV$
 - e.g. $v = -4V_T = -100mV$
 - $\Rightarrow e^{\frac{v}{V_T}} - 1 = e^{-4} - 1 = 0.018316 - 1$
- which clearly indicate that
 - exp term is very small with respect to 1
 - and can be neglected
- $\Rightarrow i = I_S \left(e^{\frac{v}{V_T}} - 1 \right) \approx -I_S$ for the reverse bias region
- $i = I_S \left(e^{\frac{v}{V_T}} - 1 \right) \approx -I_S$
 - Real diodes have reverse currents much higher than I_S
 - e.g. for a small signal diode with $I_S = 10^{-14}A$ to $10^{-15}A$
 - reverse current is on the order of $1nA$
 - Still the reverse current is very small
 - and the diode can be considered as cut off
 - in the reverse-bias region of operation.
 - the reverse current also changes with the change in temperature
 - As a rule of thumb, the reverse current doubles for
 - every $10^\circ C$ rise in temperature

The Breakdown Region



● **Figure 4.8** The diode i - v relationship with some scales expanded and others compressed in order to reveal details.

- The breakdown region is entered when
 - magnitude of the reverse voltage exceeds a certain threshold value,
 - called the breakdown voltage, V_{ZK}
- Breakdown voltage V_{ZK} is specific to the particular diode
 - and is typically in the range $2V \leq V_{ZK} \leq 2000V$
 - V_{ZK} is the voltage at the knee of the i - v curve
 - The subscript Z stands for zener and K stands for knee.

- In the breakdown region,
 - the reverse current increases rapidly,
 - while the associated increase in voltage drop is very small.
 - thus the diode i-v characteristics in breakdown
 - is almost a vertical line.
 - this region can be used for voltage regulation.
-